SPINNAKER ANALYTICS

Topic:-Weather Classification

Using Machine Learning

Exploratory Data Analysis, Feature Engineering & Model Comparison

Kirtan Patel

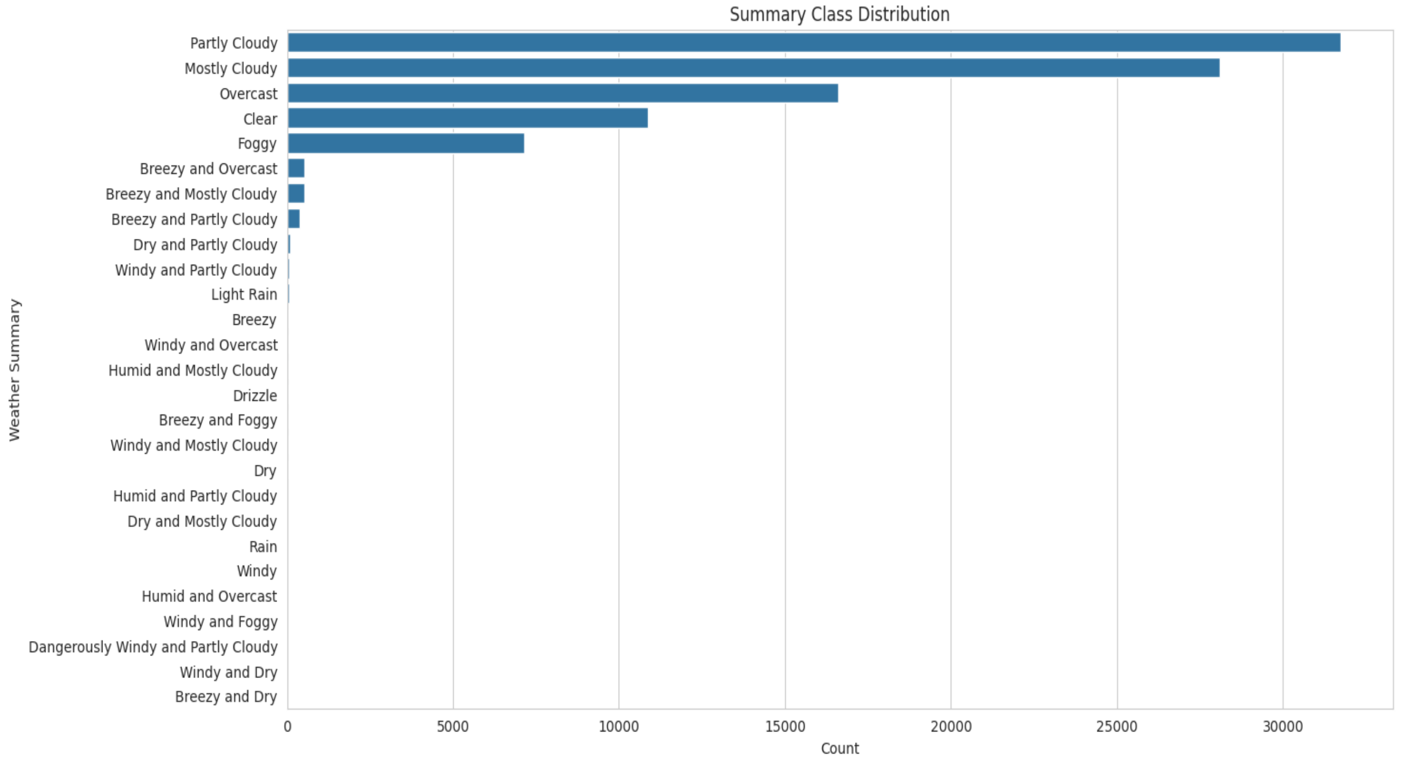
Dataset Overview

96,453	11	2006 - 2016	27
Total Rows	Features	Date Range	Unique Summaries

- Historical weather data with hourly observations
- Features: Temperature, Humidity, Wind Speed, Pressure, Visibility
- Target: Weather Summary (27 unique classes → grouped to 5 categories)
- 517 missing values in Precip Type (filled with mode)
- 24 duplicate rows removed

EDA: Weather Summary Distribution

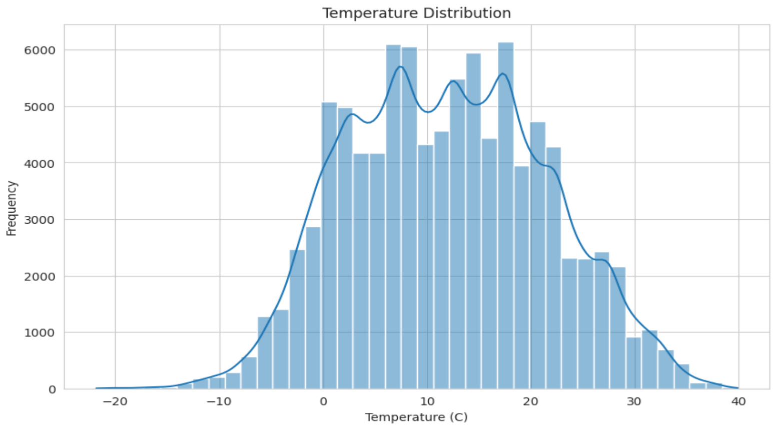
Distribution of weather summary classes in the dataset

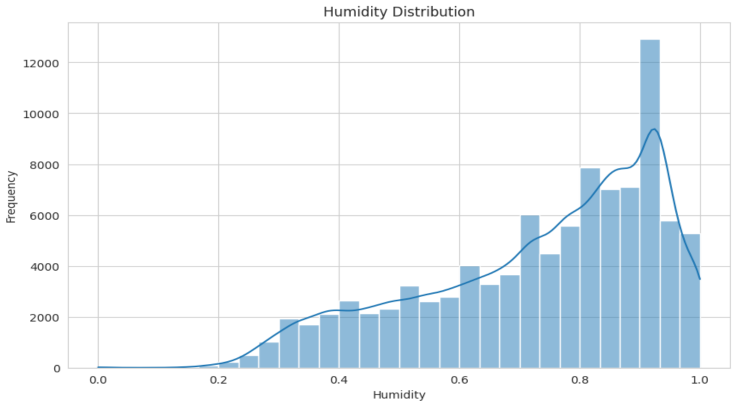


Highly imbalanced classes. A few conditions dominate while many have very low frequencies.

EDA: Temperature Distribution & Humidity Distribution

Histogram of temperature and humidity values with KDE overlay



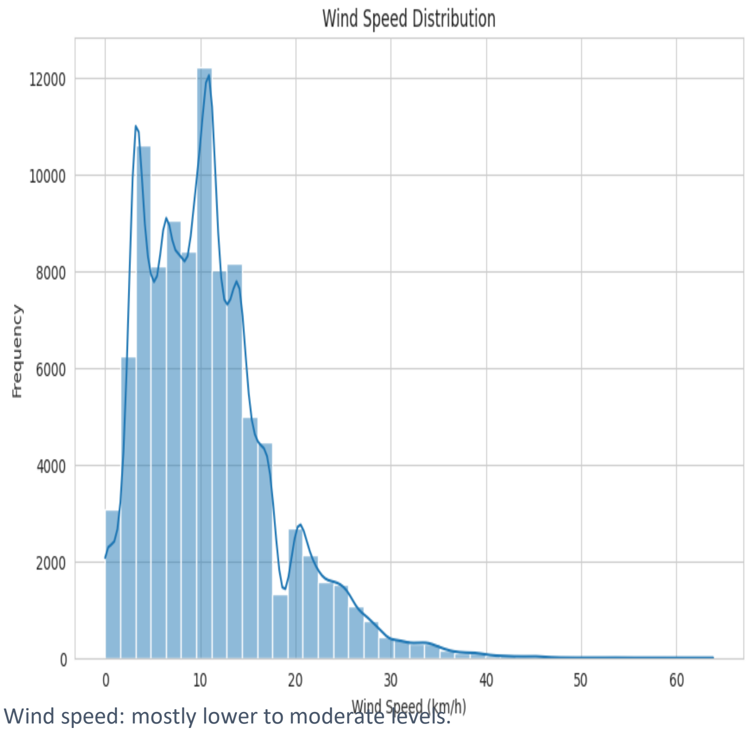


Most values concentrated between 0°C and 25°C. Extreme temperatures are rare.

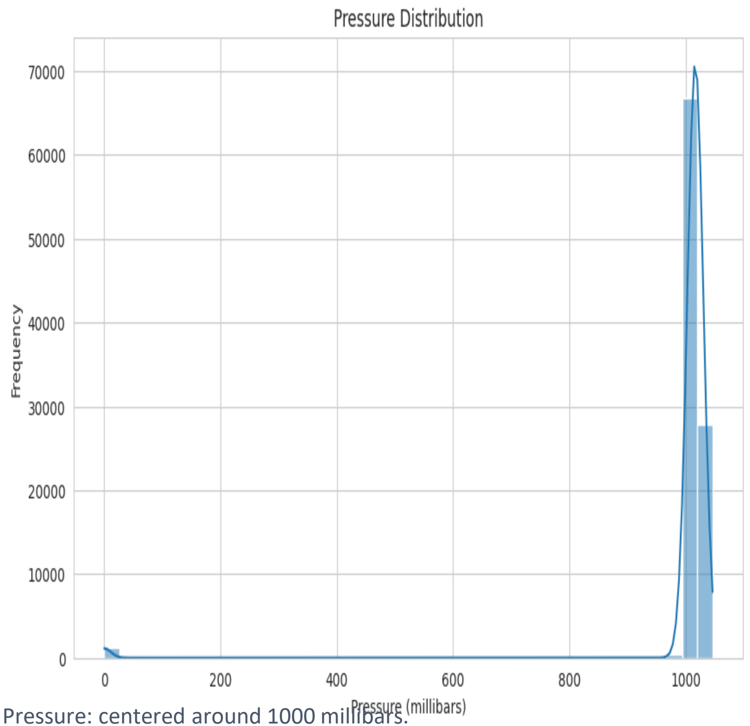
Humidity concentrated at higher levels, indicating predominantly humid conditions.

EDA: Wind Speed & Pressure

Distributions of wind speed and atmospheric pressure



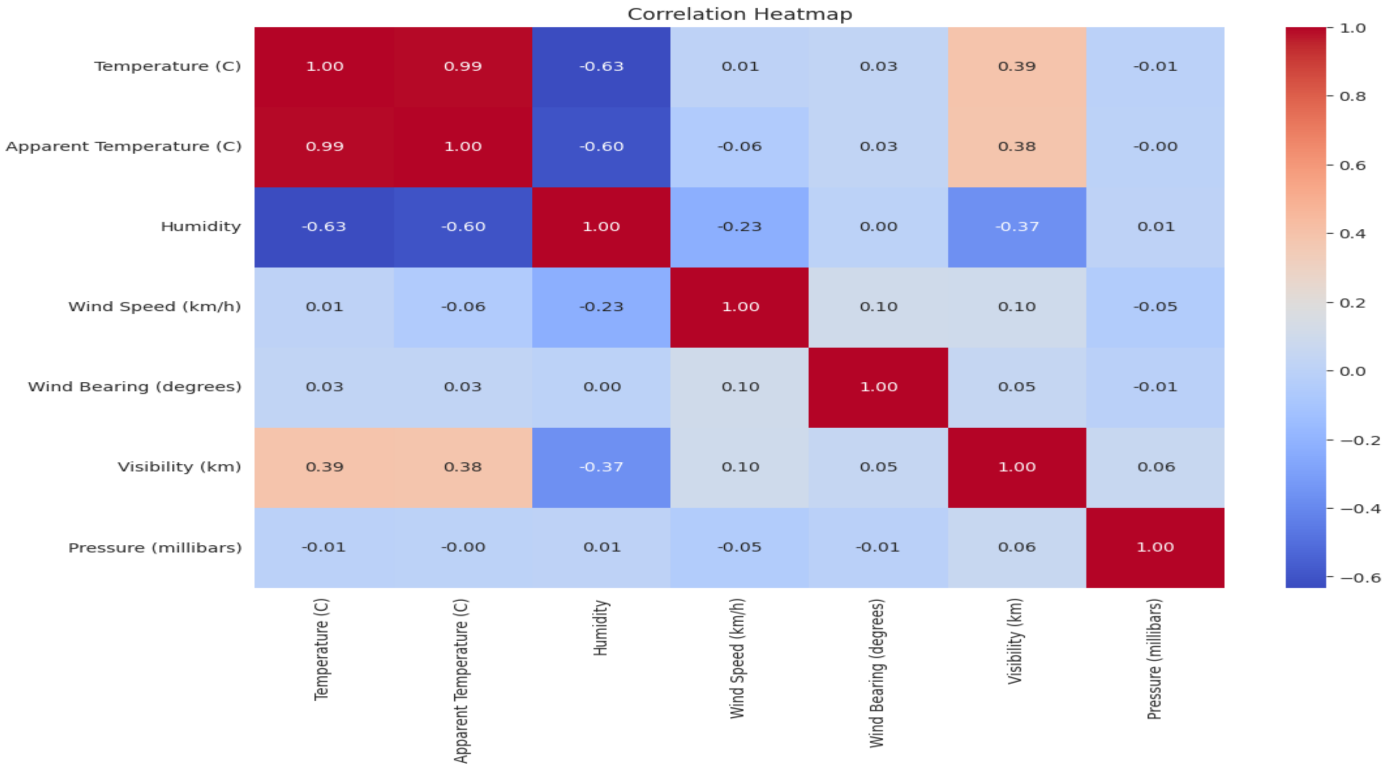
Wind speed: mostly lower to moderate levels.



Pressure: centered around 1000 millibars.

EDA: Feature Correlation

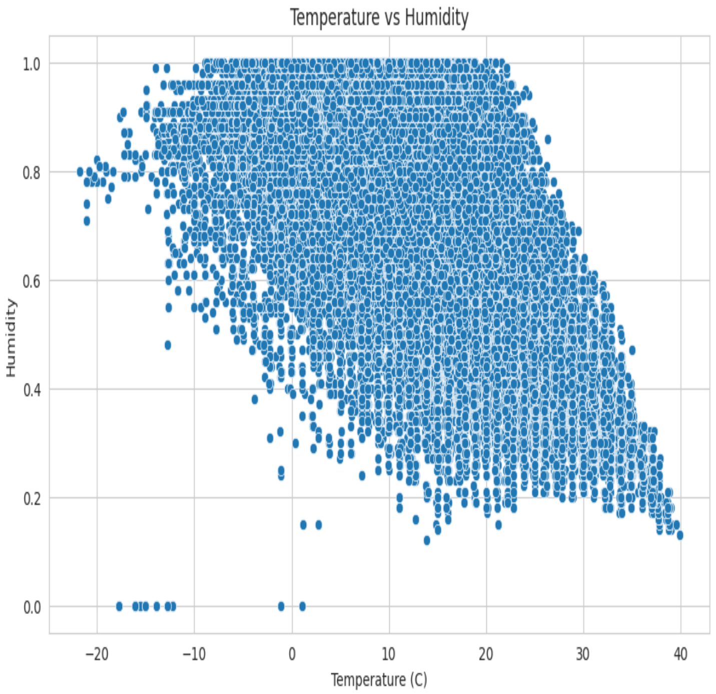
Heatmap showing pairwise correlations between numerical features



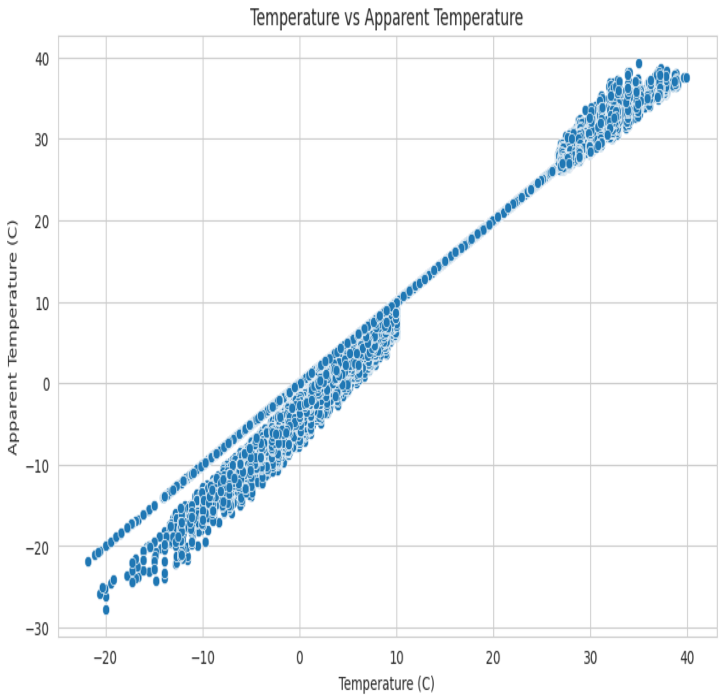
Temperature & Apparent Temperature: very strong positive correlation. Humidity has moderate negative correlation with Temperature.

EDA: Temperature Relationships

Scatter plots showing temperature vs humidity and vs apparent temperature



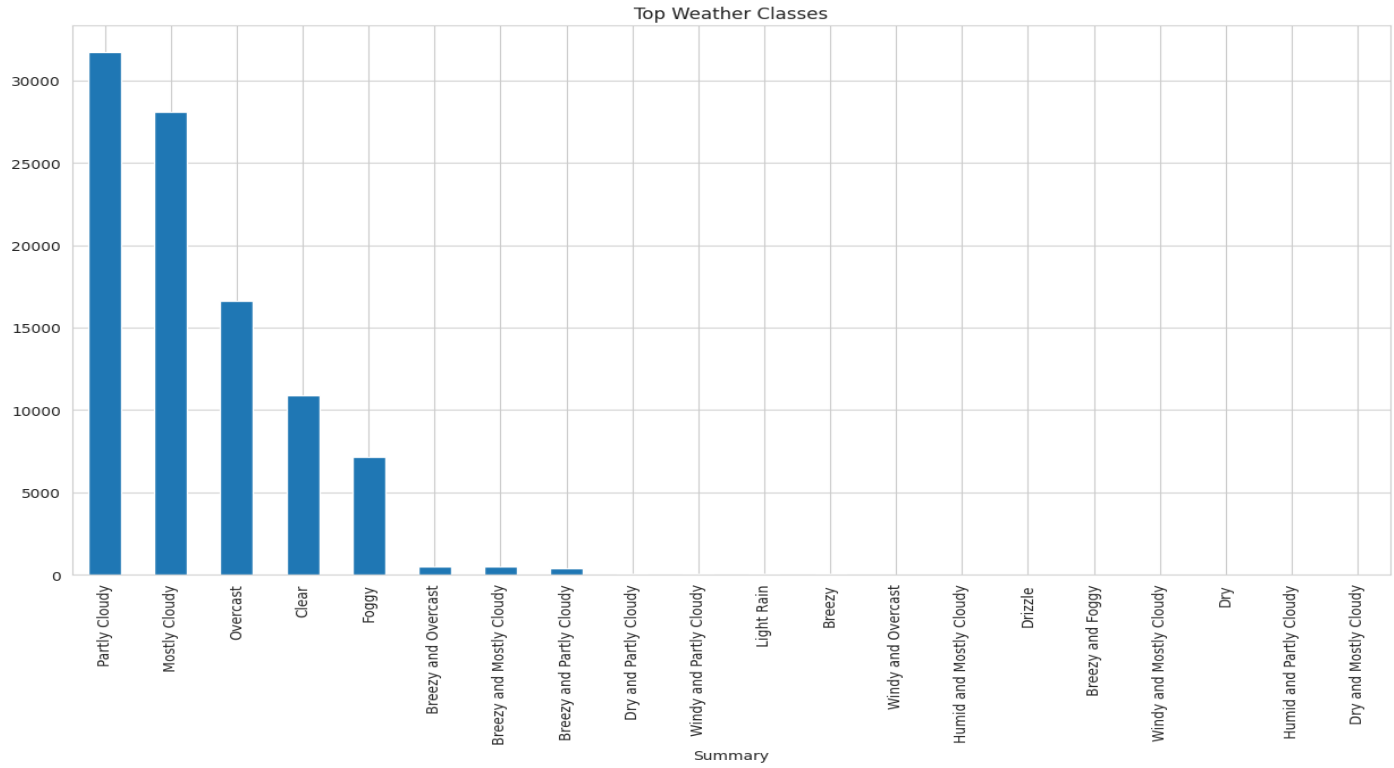
Negative relationship: humidity decreases as temperature increases.



Strong positive relationship between temperature and apparent temperature.

EDA: Top Weather Classes

Bar chart of the 20 most frequent weather summary classes



Partly Cloudy, Mostly Cloudy, Overcast, Clear, and Foggy dominate. Significant class imbalance.

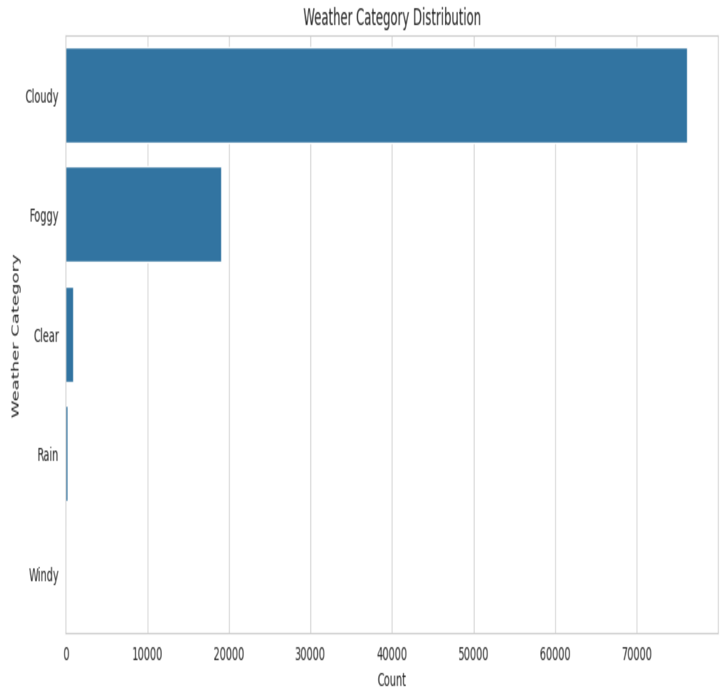
Class Grouping Strategy

The original dataset has highly fragmented, overlapping weather labels causing severe class imbalance. Grouping similar classes reduces noise and helps models learn more generalized patterns.

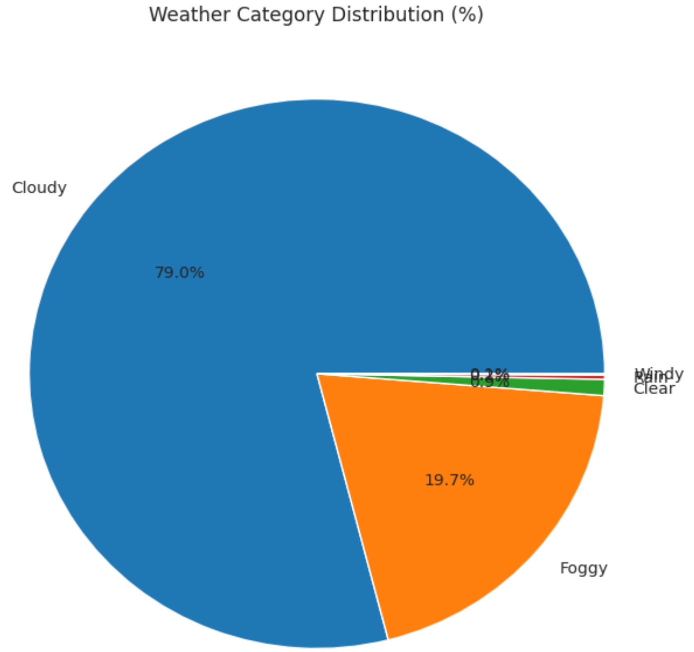
Cloudy	Partly Cloudy, Mostly Cloudy, Overcast, and all cloudy variants
Rain	Rain, Drizzle and all rainy daily summaries
Snow	All snow-related daily summaries
Windy	Windy, Breezy and wind-dominant conditions
Fog	Foggy conditions — highest feature importance in XGBoost
Clear	Clear throughout the day and clear-starting conditions

Weather Category Analysis

Grouped weather categories: count plot and pie chart



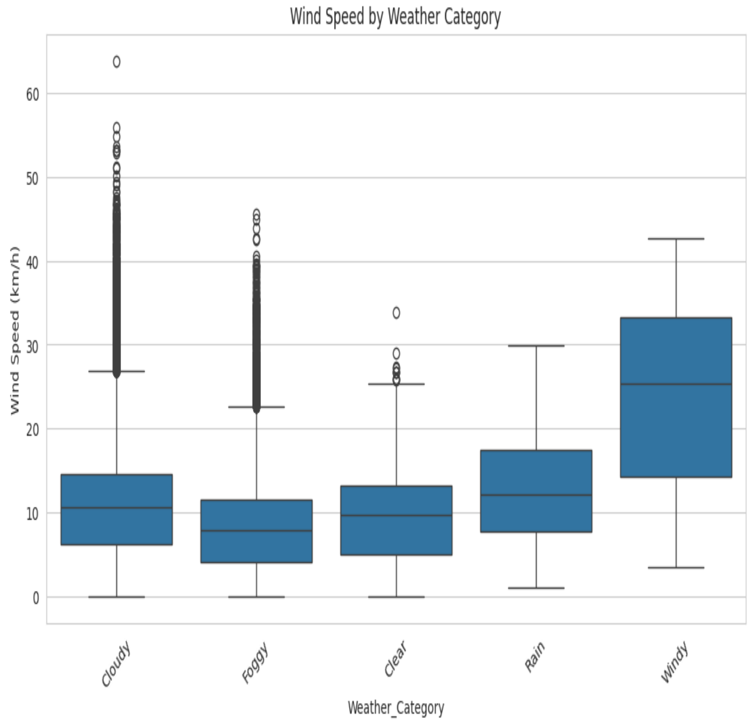
Cloudy dominates, followed by Foggy.



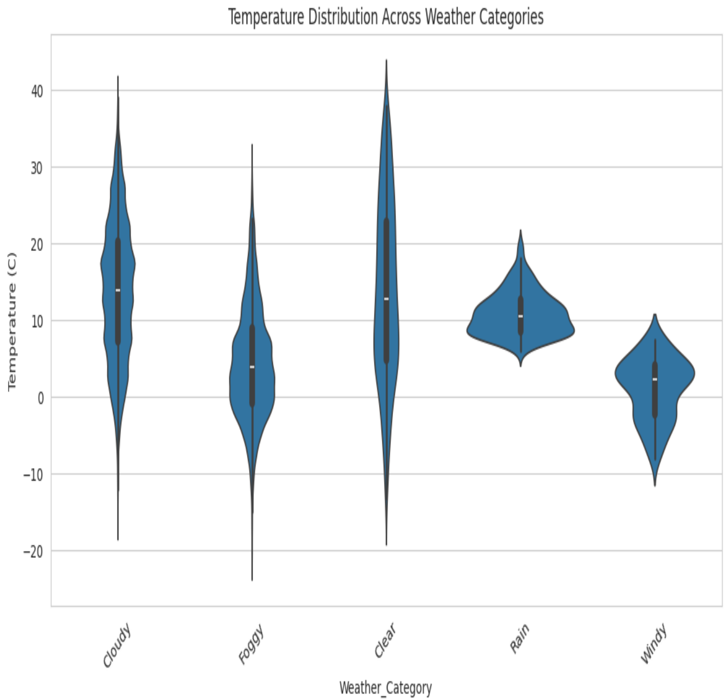
Cloudy: 79%, Foggy: 19.7%. Remaining categories are small.

Category: Wind Speed & Temperature

Wind speed boxplot and temperature violin plot by category



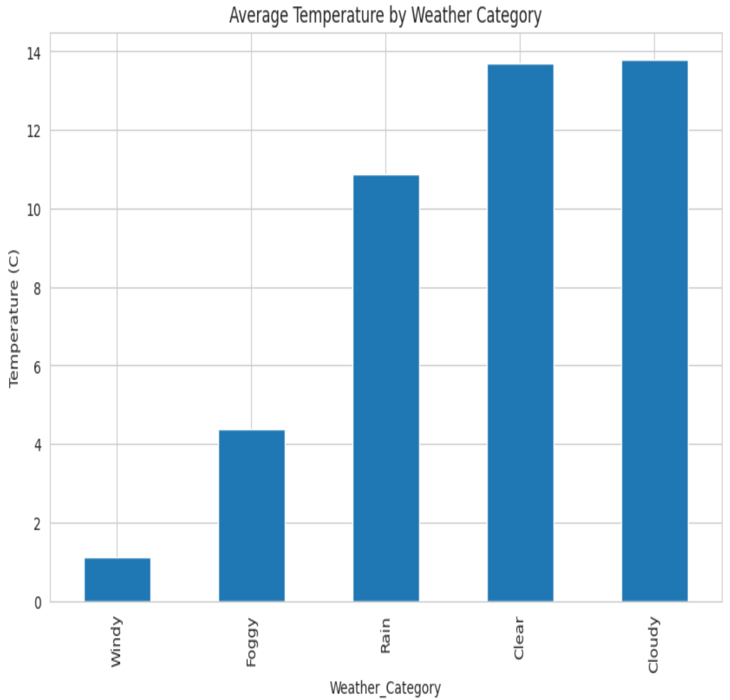
Windy conditions have significantly higher wind speeds.



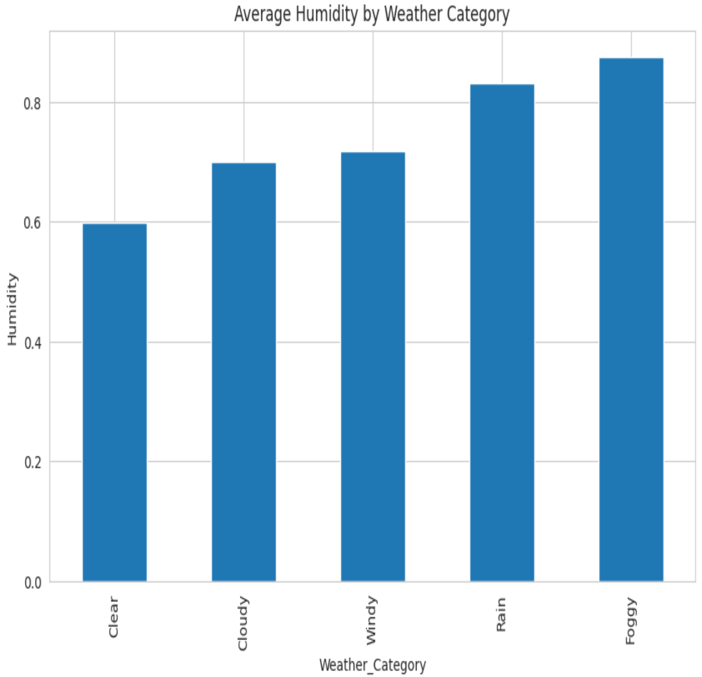
Clear & Cloudy show greater temperature variability.

Category: Averages

Average temperature and humidity by weather category



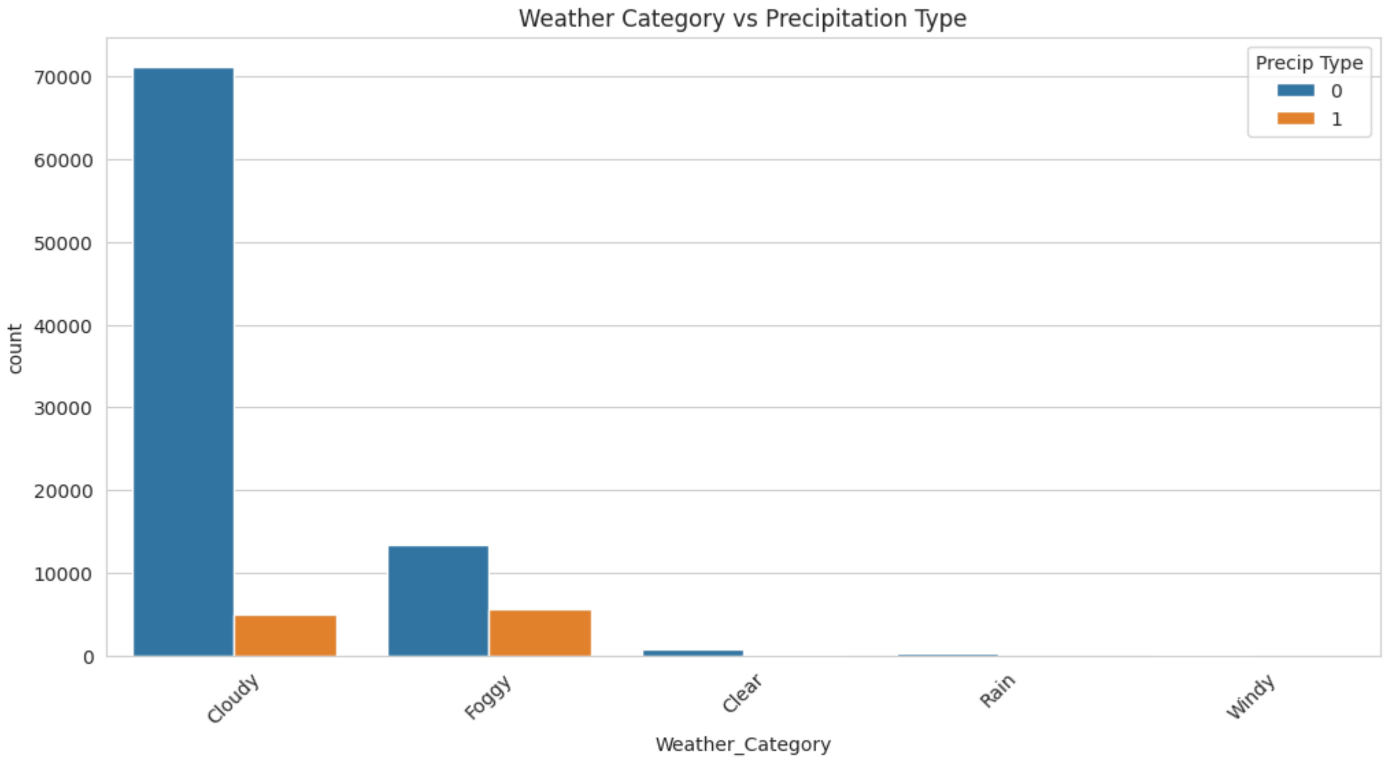
Cloudy & Clear conditions have higher average temperatures.



Foggy & Rainy conditions exhibit higher average humidity.

Category: Precipitation Type

Weather category vs precipitation type distribution



Most categories are associated with rain. Snow occurs less frequently.

Model Selection

Logistic Regression

Simple and effective baseline classification model

Linear SVM

Finds optimal decision boundaries between classes

Random Forest

Handles complex patterns, reduces overfitting via ensemble

XGBoost

High accuracy with gradient boosting, strong predictive performance

Model Performance: Before Grouping

Models trained on original 27 weather summary classes

Model	Accuracy	Macro Precision	Macro Recall	Macro F1	Weighted F1
XGBoost	81.89%	92.47%	85.16%	87.98%	81.89%
Random Forest	59.70%	80.99%	44.86%	54.19%	57.99%
Logistic Regression	27.14%	20.19%	15.44%	15.76%	18.77%
SVM	25.47%	13.67%	11.54%	10.82%	15.02%

- 27 original classes caused severe class imbalance — many classes had very few samples
- XGBoost performed best (81.89% accuracy) but still struggled with minority classes
- Logistic Regression and SVM performed poorly (~25-27%) — unable to handle the high number of classes
- Low macro recall scores indicate models failed to predict many minority classes
- This motivated grouping similar weather summaries into 5 broader categories

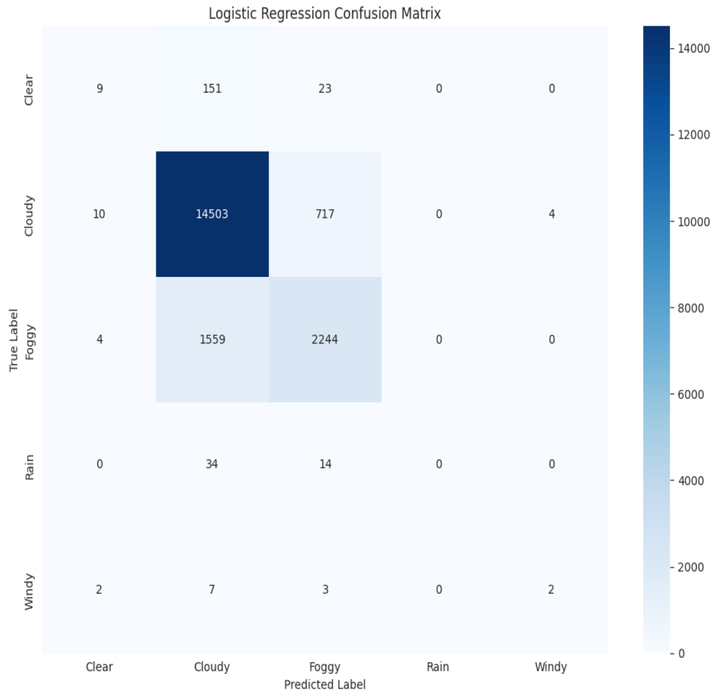
Model Performance Summary

Model	Accuracy	Precision	Recall	F1 Score
XGBoost	95.39%	95.33%	95.39%	95.28%
Random Forest	91.52%	91.44%	91.52%	90.91%
SVM	86.96%	85.21%	86.96%	85.77%
Logistic Regression	86.89%	85.61%	86.89%	85.87%

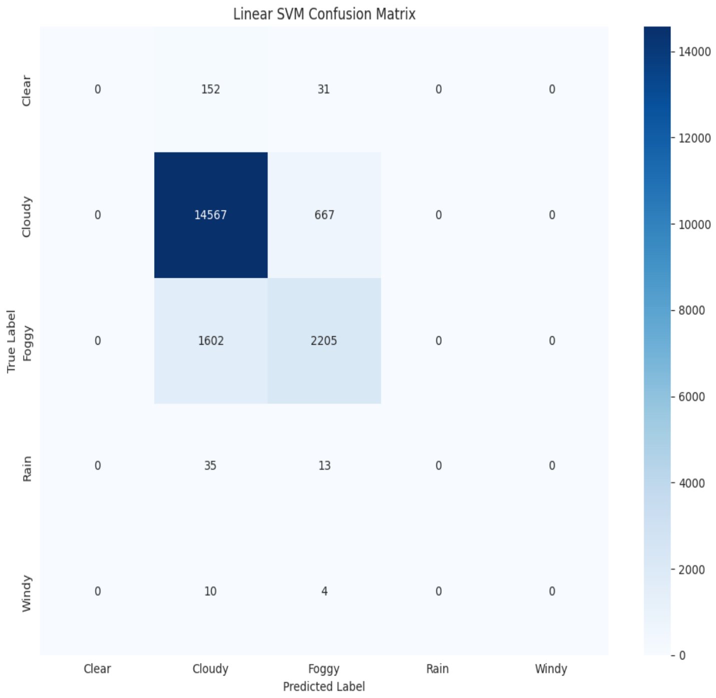
- XGBoost achieves the best performance across all metrics
- Random Forest is a strong second, especially for minority classes
- Logistic Regression and SVM perform similarly as baseline models
- All models benefit significantly from weather category grouping (reduced from 27 to 5 classes)

Model Evaluation: LR & SVM

Confusion matrices for Logistic Regression and Linear SVM (after grouping)



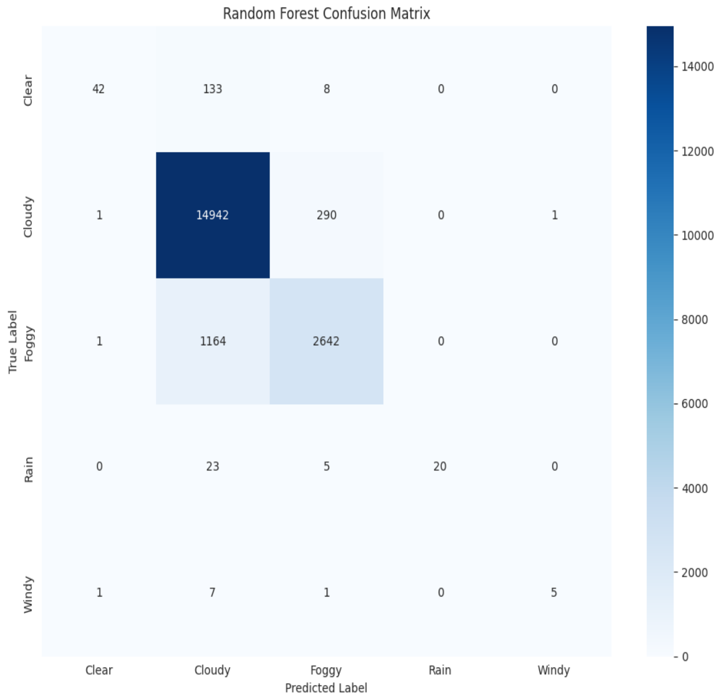
LR: Accuracy 86.9%. Strong on Cloudy (95% recall). Weak on minority classes.



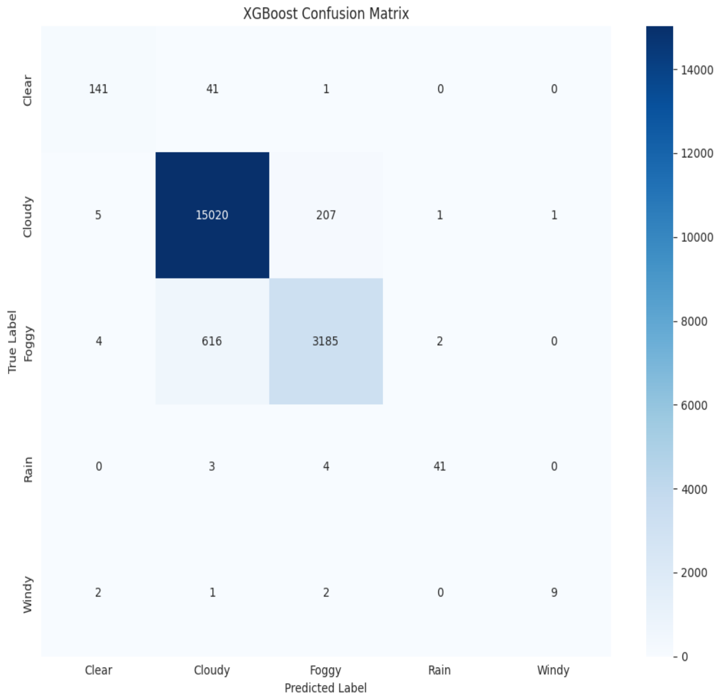
SVM: Accuracy 87.0%. Similar pattern. Fails on Clear, Rain, Windy.

Model Evaluation: RF & XGBoost

Confusion matrices for Random Forest and XGBoost (after grouping)



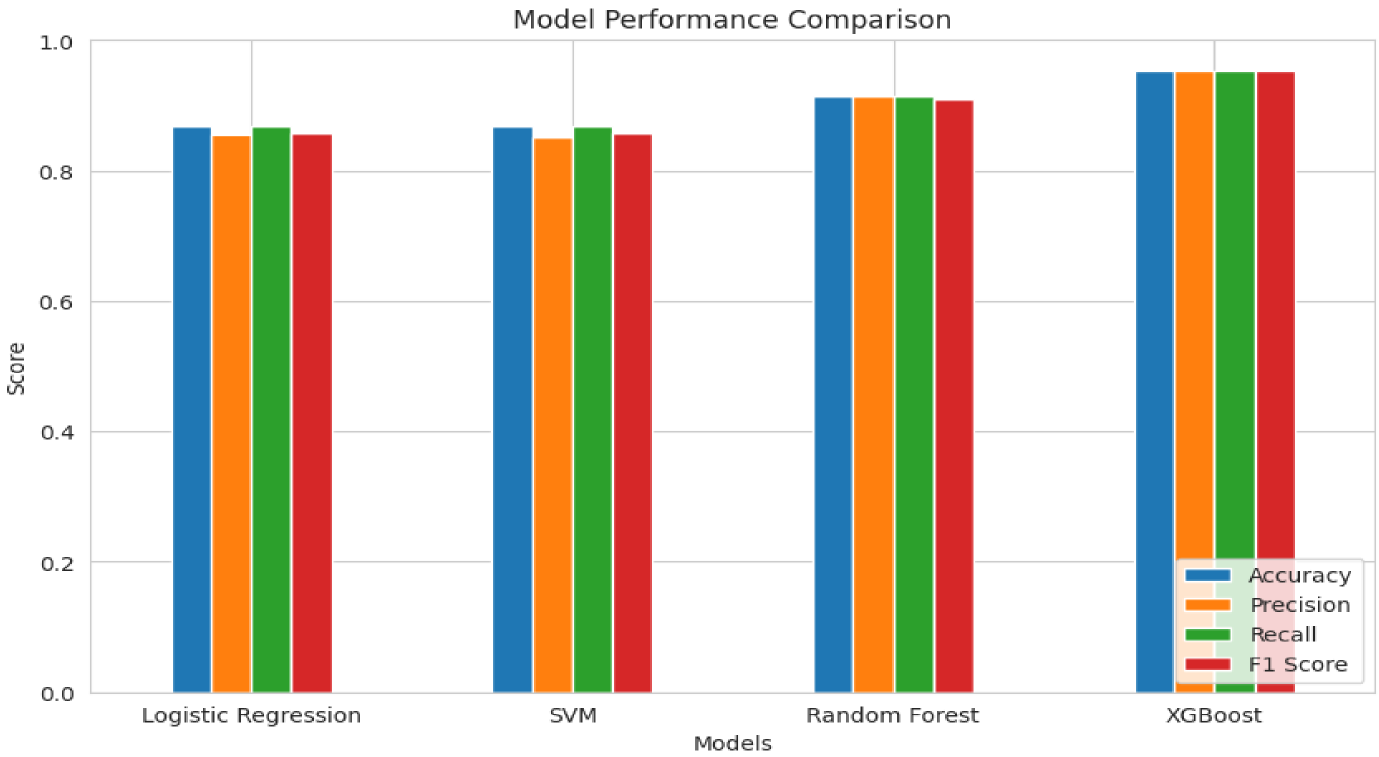
RF: Accuracy 91.5%. Better on minority classes than LR/SVM.



XGBoost: Accuracy 95.4%. Best overall performance across all classes.

Model Comparison

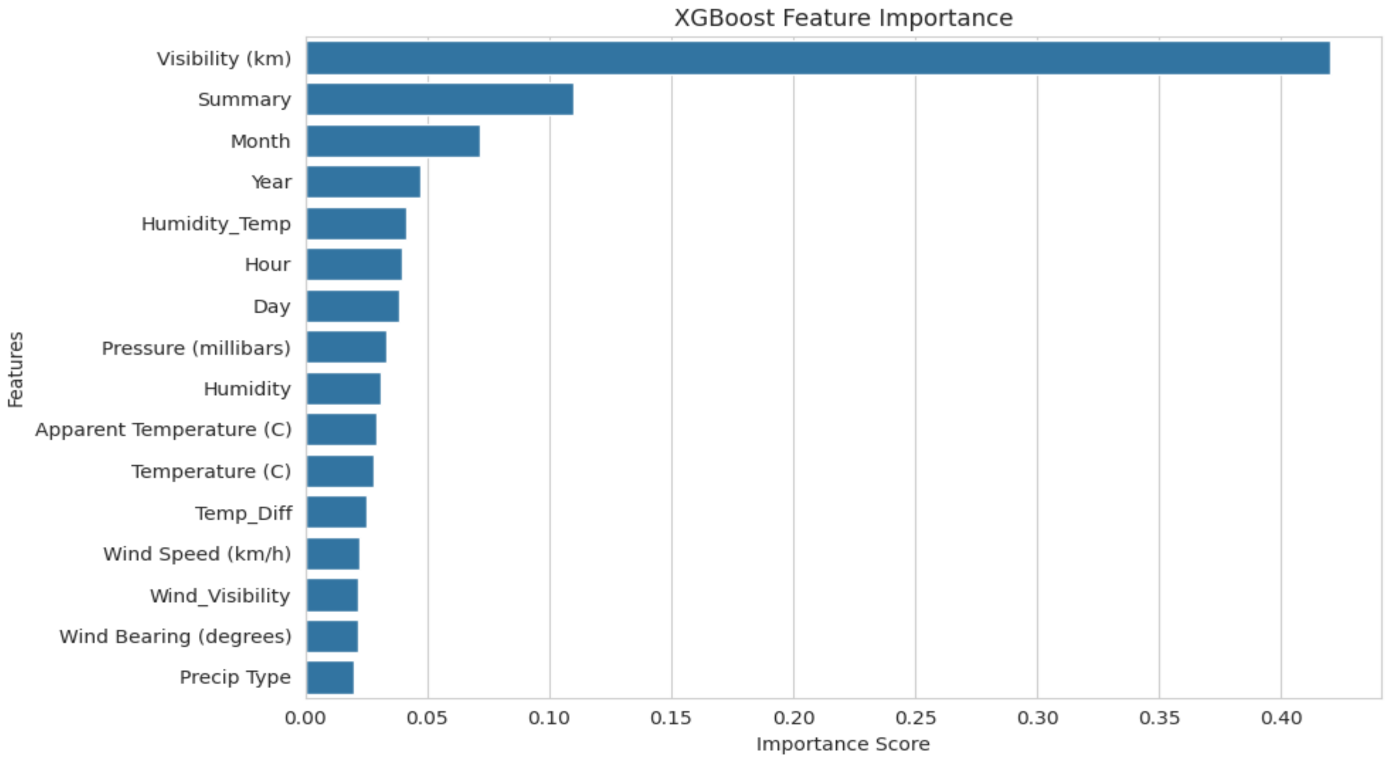
Side-by-side comparison of all four models on Accuracy, Precision, Recall, and F1 Score



XGBoost leads with 95.4% accuracy, followed by Random Forest at 91.5%. LR and SVM perform similarly at ~87%.

Feature Importance (XGBoost)

XGBoost feature importance scores showing contribution of each feature



Visibility is the most important feature (42%), followed by Summary (11%) and Month (7%).

Conclusion

- Successfully developed a weather classification system using ML on historical data (96,453 records, 2006-2016)
- Comprehensive EDA revealed feature distributions, correlations, and class imbalance in weather summaries
- Weather category grouping (27 → 5 classes) significantly improved model performance
- XGBoost achieved the best results: 95.4% accuracy, 95.3% precision, 95.3% recall, 95.3% F1
- Visibility (42%) and Summary (11%) are the most important predictive features
- Feature engineering (Temp_Diff, Humidity_Temp, time features) enhanced model inputs
- Tree-based ensemble methods (RF, XGBoost) substantially outperform linear models (LR, SVM)

Thank You